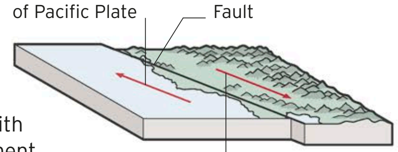
The San Andreas Fault

Part of the boundary along which two tectonic plates grind past each other, producing frequent earthquakes

The San Andreas Fault is one of few places on Earth where the boundary between two tectonic plates can be seen running across land. Extending for approximately 810 miles (1,300 km) through California, the San Andreas is known as a continental transform fault. Along the fault, the Pacific Plate is slowly slipping past the North American Plate (see panel, below). Each slippage—which typically occurs only along a section of the fault—is accompanied by an earthquake. A large tremor that devastated San Francisco in 1906 is the most famous quake associated with movements on the fault.

MOVEMENT ALONG THE FAULT

The Pacific and North American plates are moving about $1\frac{3}{4}$ in (4.6 cm) per year in relation to each other. This motion results in jerky movement on the fault itself, with periods of little or no displacement interspersed with sudden big slippages during major earthquakes.



△ **LINEAR SCAR**
In parts of California, the San Andreas Fault visibly bisects the landscape, as seen here in the Carrizo Plains, around 70 miles (120 km) northwest of Los Angeles.

△ **COLORPLAY**
Grand Prismatic Spring is the largest hot spring both in Yellowstone and in the U.S. Its colors come from pigments produced by heat-loving microbes in the water.

▷ **SCALDING HOT**
The temperatures of these hot springs can be close to boiling point, which means they continually give off water vapor. This turns into dramatic plumes of steam when the air temperature above the spring's surface cools.